

RWA Tokenization Platform Audit Report

1. Executive Summary

The RWA Tokenization Platform is a modular decentralized finance protocol developed using Solidity and the Foundry framework.

The protocol combines:

- ERC20 governance token
- ERC4626 tokenized vault
- DAO governance
- Chainlink oracle integration
- Upgradeable smart contracts
- Automated market maker (AMM)
- Security testing infrastructure
- CI/CD automation

The primary objective of this audit was to evaluate:

- smart contract security
- access control mechanisms
- upgradeability safety
- governance protections
- oracle security
- vault accounting consistency
- AMM invariant preservation
- deployment reliability

The audit process included:

- manual review
- unit testing
- fuzz testing
- invariant testing
- fork testing
- Slither static analysis
- CI/CD validation

2. Audit Scope

Included Contracts

Contract	Description
RWAToken	Governance ERC20 token
RWAVault	ERC4626 vault
Governance	DAO governance system
OracleAdapter	Chainlink oracle integration
RWAFactory	CREATE2 deployment factory
RWATokenUpgradeableV1	UUPS upgradeable token
RWATokenUpgradeableV2	Upgraded implementation
SimpleAMM	Automated market maker

3. Security Methodology: The audit used multiple security analysis methodologies.

Manual Review

Manual analysis included:

- access control validation
- privilege escalation checks
- governance logic review
- vault accounting review
- oracle freshness validation
- upgrade authorization checks

Automated Testing

Testing methodologies included:

- unit testing
- fuzz testing
- invariant testing
- fork testing

Static Analysis

Static analysis was performed using:

- Slither

Continuous Integration

Automated CI pipelines validated:

- builds
- tests
- coverage
- deployment consistency

4. RWAToken Security Analysis

Overview

RWAToken is implemented using:

- ERC20Votes
- ERC20Permit
- AccessControl

The token supports:

- governance voting
- delegation
- vote snapshots
- minting
- burning

Security Review

Access Control: Minting functionality is protected using role-based access control.

Only authorized accounts may:

- mint tokens
- manage governance permissions
- update protocol configuration

Governance Security

Snapshot voting prevents flash-loan governance attacks.

Voting power is calculated using historical checkpoints.

Permit Signatures

ERC20Permit support enables gas-efficient approvals while maintaining signature security.

5. RWAVault Security Analysis

Overview

RWAVault implements the ERC4626 vault standard.

Core functionality includes:

- deposits
- withdrawals
- share minting
- share redemption
- asset/share conversion

Security Review

Accounting Consistency

The vault preserves asset/share consistency during deposits and withdrawals.

Preview Functions

Preview calculations correctly estimate:

- shares minted
- assets withdrawn
- conversion ratios

ERC4626 Compliance

The implementation follows ERC4626 standards for interoperability and predictable vault behavior.

6. Governance Security Analysis

Governance Architecture

The governance system uses:

- OpenZeppelin Governor
- TimelockController
- ERC20Votes

Governance lifecycle:

1. Proposal creation
2. Voting delay
3. Voting period
4. Queue
5. Timelock delay
6. Execute

Security Protections

Timelock Delay

Timelock delays prevent instant execution of malicious proposals.

Quorum Protection

Quorum requirements reduce governance manipulation risks.

Proposal Threshold

Proposal thresholds reduce governance spam.

Snapshot Voting

Snapshot voting prevents temporary voting power manipulation.

7. Oracle Security Analysis

OracleAdapter

OracleAdapter integrates Chainlink price feeds.

Security Features

- stale price protection
- timestamp validation
- controlled feed updates
- freshness verification

Findings

The oracle implementation correctly rejects stale price data.

Feed updates are restricted to authorized accounts.

8. Upgradeability Security Analysis

UUPS Proxy Pattern

The protocol includes upgradeable contracts using the UUPS architecture.

Upgradeable components:

- RWATokenUpgradeableV1
- RWATokenUpgradeableV2

Security Review

Upgrade Authorization

Upgrade functionality is protected using authorization controls.

Storage Persistence

Storage variables remain consistent across upgrades.

Initialization Protection

Initialization logic prevents unauthorized reinitialization.

9. AMM Security Analysis

Overview

SimpleAMM implements a constant product automated market maker.

Formula used:

$$x * y = k$$

Core Features

- liquidity provision
- liquidity withdrawal
- token swaps
- reserve accounting

Security Review

Invariant Preservation

The AMM correctly preserves reserve invariants during swaps.

Liquidity Accounting

LP share accounting remains consistent after deposits and withdrawals.

Swap Validation

Reserve updates correctly maintain pricing consistency.

10. Reentrancy Analysis

Reentrancy Testing

The project includes:

- vulnerable contract examples
- secure implementations
- attack simulations
- reentrancy protection tests

Security Measures

Secure contracts follow:

- Checks-Effects-Interactions
- proper state update ordering

- controlled external calls

Result

No successful reentrancy exploits were identified in secure implementations.

11. Testing Results

Unit Testing

The protocol includes comprehensive unit tests for:

- RWAToken
- RWAVault
- Governance
- OracleAdapter
- Factory
- Upgradeability
- AMM functionality

Fuzz Testing

Fuzz testing validated:

- randomized transfers
- vault accounting
- governance safety
- swap calculations

Invariant Testing

Invariant tests validated:

- total supply consistency
- treasury solvency
- vault accounting invariants
- protocol stability

Fork Testing

Fork tests interacted with:

- USDC
- Uniswap V2
- Chainlink feeds

This improved integration confidence under real network conditions.

12. Coverage Analysis

Coverage metrics achieved:

Metric	Result
Line Coverage	94.21%
Statement Coverage	93.33%
Function Coverage	92.31%

Coverage analysis demonstrates strong automated testing coverage across the protocol.

13. Slither Static Analysis

Static analysis was performed using Slither.

Analysis included:

- reentrancy detection
- dead code analysis
- shadowing checks
- unused variable detection
- access control review

No critical vulnerabilities were identified during static analysis.

14. CI/CD Validation

GitHub Actions workflows automatically execute:

- forge build
- forge test

- forge coverage
- Slither analysis

CI/CD automation improves:

- deployment safety
- testing reliability
- code consistency
- protocol maintainability

15. Deployment Review

Deployment scripts support Base Sepolia deployment.

Deployment process includes:

- governance deployment
- vault deployment
- proxy deployment
- AMM deployment

Environment configuration uses:

- RPC URLs
- private keys
- environment variables

16. Findings Summary

Severity	Findings
Critical	0
High	0
Medium	0

Low	Minor optimization opportunities
Informational	Gas optimization recommendations

17. Recommendations

Recommended future improvements:

- multi-oracle aggregation
- emergency pause mechanism
- advanced AMM pricing
- frontend monitoring dashboard
- analytics integration
- cross-chain deployment support

18. Conclusion

The RWA Tokenization Platform demonstrates a modular and security-oriented architecture using modern Solidity engineering practices.

The protocol successfully integrates:

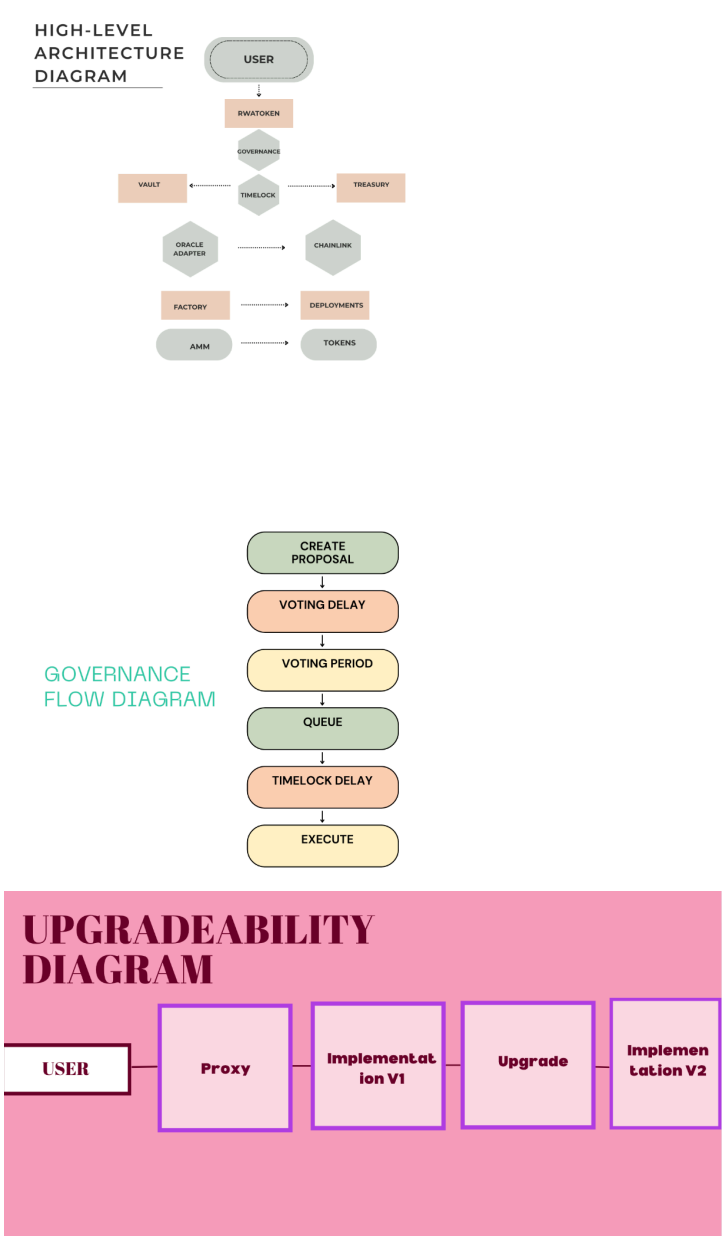
- governance
- tokenization
- upgradeability
- oracle infrastructure
- DeFi liquidity mechanisms
- advanced testing methodologies

The audit identified no critical security vulnerabilities.

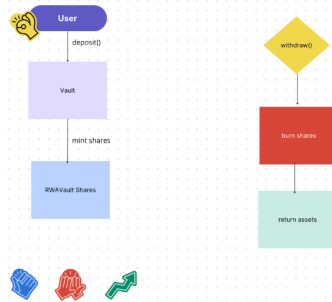
The project demonstrates strong engineering practices including:

- modular architecture
- extensive testing
- CI/CD automation
- upgradeable infrastructure
- access control protections
- oracle validation
- governance safety mechanisms

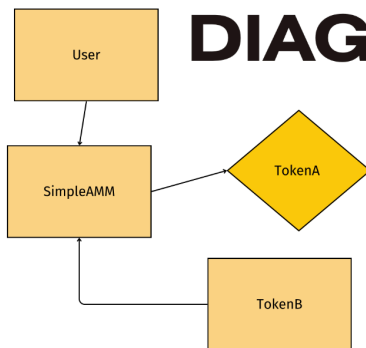
Overall, the protocol reflects production-style blockchain development practices and secure smart contract engineering principles.



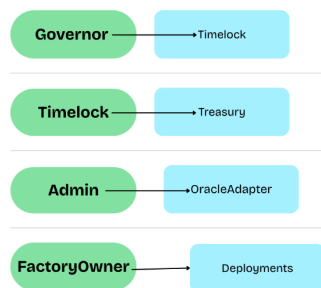
VAULT FLOW DIAGRAM



AMM FLOW DIAGRAM



Data Flow Diagram



TOKENIZE • GOVERN • EARN

The future of RWA governance

A decentralized platform for tokenized real-world assets, vault deposits, DAO governance, and onchain voting.

Connected Wallet
Not Connected

Base Sepolia

Token Balance

0

Voting Power

0

Delegate Address

No Delegate

Delegate address...

Delegate Votes

Vault Deposit

DAO GOVERNANCE

Governance Portal

\$20,000 for something

Create Proposal

Indexed Governance Proposals

Proposal #116216690472...
Aaa kopen

View Governance Contract →
0x5022...a2a9

Proposal #108938110545...
Frontend Proposal 1778926960525

View Governance Contract →
0x5022...a2a9

Proposal #582744397778...
Frontend Proposal 1778926845556

View Governance Contract →
0x5022...a2a9

Proposal #110253596831...
Frontend Proposal 1778923738376

View Governance Contract →
0x5022...a2a9

Proposal #566564248669...
Frontend Proposal 1778923204363

View Governance Contract →
0x5022...a2a9

Recent Governance Votes

Aaa kopen
Support: FOR

Weight: 1000000
0x5022...a2a9

Frontend Proposal 1778926960525
Support: FOR

Weight: 1000000
0x5022...a2a9

Proposal ID

Paste proposal ID ...

Check Proposal State

Current State

Cast your governance vote

Vote FOR

Vote AGAINST

Network

Base Sepolia

